BBF 302E Database Systems
Term Project Proposal

Milestone 1 · Fall 2026

1. Team Information

Team name	Hanabi
Selected project topic	Board Game Community & Game Night
Topic registered in the mapping document?	Yes
Course section(s)	Fatuma Zehra Erdem - 12495 Şevval Tüfekçi - 12495 Sudenaz Hasret - 12492 Mahmut Kartal - 12492 Fikriye Zeynep Battal - 15742

#	Full name	Student ID	Section	Role in the team	E-mail (ITU)
1	Fatuma Zehra Erdem	150230102	12495	Team Leader – Game Catalog & Game Libraries	erdemf23@itu.edu.tr
2	Şevval Tüfekçi	150230079	12495	Accounts, Profiles & Host Ratings	tufekci23@itu.edu.tr
3	Sudenaz Hasret	150230072	12492	Game Nights & Sessions	hasret23@itu.edu.tr
4	Mahmut Kartal	150240114	12492	Participation, Reservations & Check-in	kartalm24@itu.edu.tr
5	Fikriye Zeynep Battal	150240106	15742	Scoring, Leaderboard & Player Reports	battalf24@itu.edu.tr

2. Project Overview

2.1 Project description

Board Game Community & Game Night Organizer is a web-based community platform that brings board game players together in real life. Members register the board games they own, organize game nights and join game nights organized by others. A host creates a game night with a date, time, venue and total capacity, and can optionally restrict participation by age range, gender or experience level. Within a night, the host and co-hosts open one or more game sessions (in parallel or one after another, and new ones can be opened during the night); each session has its own game, start time, player capacity and the person who brings the game. Participants join a night and reserve seats only in the sessions they want to play.
During play, the host records and updates in-game scores through the application. When the host confirms the final scores and winners, leaderboard points are awarded; points are weighted by the game's typical play time (e.g., winning Monopoly is worth far more than winning UNO). Users can see their game history, results, total points, per-game rankings and attendance streaks, rate the host after a night and report problematic players. Every member has a profile with a Player side (photo, bio, total score) and, if hosting is enabled, a

Host side (owned games, hosted nights, host rating). A public game catalog lists every game in the system, so anyone curious about a game can browse it before buying. The target users are board game enthusiasts, local gaming groups and board game cafés.

2.2 Project scope

- Member registration, login and profiles (player side and host side).
- Public game catalog and personal game libraries of hosts.
- Creating, editing and cancelling game nights with participation conditions and capacity.
- Co-host management (venue, games and session responsibilities).
- Game sessions inside a night, seat reservations and check-in of players who actually play.
- Live score entry, result finalization, weighted leaderboard points, game history, per-game rankings and attendance streaks.
- Host ratings, player reports and admin moderation.

3. User Roles

Role	Description	What this role can do
Guest	Unauthenticated visitor	Browse the game catalog and public list of upcoming game nights; register or log in.
Member (Player)	Registered user; every user is a member and acts as a player by default	Manage own profile; join/leave game nights; reserve/cancel session seats; view history, points, rankings and streaks; rate hosts; report players.
Host	Member who has enabled hosting	Manage own game library; create/edit/cancel game nights and set conditions; add/remove co-hosts; create sessions; check in players; enter scores and finalize results.
Co-host	Participant promoted by the host for a specific night	Bring games; create and manage sessions of that night; check in players; enter/update scores (finalization stays with the host).
Admin	Platform moderator	Manage the game catalog (add/edit/delete games and point weights); review player reports; suspend or remove accounts.

4. Functional Requirements

ID	Requirement	User role(s)	Responsible
FR-01	A guest can register with e-mail and password, members can log in/out and edit their profile (photo, bio, birth date, gender, experience level).	Guest, Member	Şevval Tüfekçi
FR-02	A member's profile page shows a Player tab (photo, bio, total points, streak) and, if hosting is enabled, a Host tab (owned games, hosted nights, average host rating).	All	Şevval Tüfekçi

ID	Requirement	User role(s)	Responsible
FR-03	The system updates a member's attendance streak automatically when a game night they checked in to is completed, and resets it after a period of inactivity.	System	Şevval Tüfekçi
FR-04	After a game night ends, each checked-in participant can rate the host (1–5 stars with an optional comment) once per night and edit or delete their rating.	Member	Şevval Tüfekçi
FR-05	Anyone can browse and search the game catalog (name, category, player count, play time) and open a game detail page.	All	Fatuma Zehra Erdem
FR-06	A member can enable hosting and manage the games they own (add from the catalog, update condition/notes, remove).	Member, Host	Fatuma Zehra Erdem
FR-07	An admin can add, edit and delete catalog games, including the average play time that determines the game's leaderboard point weight.	Admin	Fatuma Zehra Erdem
FR-08	A game's detail page shows the per-game ranking (top players by points won in that game) and the members who own it.	All	Fatuma Zehra Erdem
FR-09	A host can create, edit and cancel a game night with date, time, venue and total participant capacity.	Host	Sudenaz Hasret
FR-10	A host can set optional participation conditions for a night (age range, gender, minimum experience level).	Host	Sudenaz Hasret
FR-11	A host or co-host can create sessions in a night by choosing a game from the host's/co-hosts' libraries, setting start time, player capacity and the person bringing the game; sessions can run in parallel or in sequence and can be added during the night.	Host, Co-host	Sudenaz Hasret
FR-12	Members can list and filter upcoming game nights (date, location, game, open seats) and view a night's sessions.	All	Mahmut Kartal
FR-13	A member can join or leave a game night; the system rejects the request if the night is full or the member does not meet its conditions.	Member	Mahmut Kartal
FR-14	A host can promote a participant to co-host (and demote them), recording their responsibilities: venue, bringing games, managing sessions. Only co-hosts may bring games.	Host	Mahmut Kartal
FR-15	A participant (including hosts and co-hosts) can reserve or cancel a seat in any session of a night they joined, without exceeding session capacity or overlapping sessions.	Member, Host, Co-host	Mahmut Kartal
FR-16	When a session starts, the host or co-host checks in the players who actually play and marks no-shows.	Host, Co-host	Mahmut Kartal
FR-17	During a session the host or co-host can add and update the in-game scores of checked-in players.	Host, Co-host	Fikriye Zeynep Battal
FR-18	At the end of a session the host confirms final scores and winners; only then are leaderboard points awarded, weighted by the game's play time, and the result is locked.	Host	Fikriye Zeynep Battal
FR-19	Members can view the global leaderboard and their own game history (sessions played, scores, wins, points earned).	Member	Fikriye Zeynep Battal

ID	Requirement	User role(s)	Responsible
FR-20	After a night, participants can report a player they had problems with; admins review reports and mark them as resolved or take action.	Member, Admin	Fikriye Zeynep Battal

5. Non-Functional Requirements

ID	Category	Requirement	Responsible
NFR-01	Security	Passwords are stored only as salted hashes (bcrypt), all pages except the catalog, night list, login and register require login, and unauthenticated requests are redirected to /login.	Şevval Tüfekçi
NFR-02	Privacy	Birth date and gender are never shown on public pages; they are used only for eligibility checks. Player reports are visible only to admins.	Şevval Tüfekçi
NFR-03	Performance	Catalog search returns results in under 500 ms for 1,000+ games, with paginated results (20 per page).	Fatuma Zehra Erdem
NFR-04	Maintainability	The Flask app is organized into one blueprint per module with raw SQL in a separate data-access layer; a requirements.txt and database creation/population scripts allow setup in under 10 minutes.	Fatuma Zehra Erdem
NFR-05	Security	All SQL statements use parameterized queries (no string concatenation) to prevent SQL injection; every form input is validated on the server side.	Sudenaz Hasret
NFR-06	Usability	Pages are rendered with Jinja templates, are usable on mobile and desktop screen widths, and show form validation errors next to the related field.	Sudenaz Hasret
NFR-07	Authorization	Night-specific actions (editing a night, managing sessions, check-in, scoring) are checked on the server against the user's role in that night; unauthorized requests receive HTTP 403.	Mahmut Kartal
NFR-08	Reliability	Joining a night and reserving a seat run in a single transaction with row locking, so capacity is never exceeded under concurrent requests.	Mahmut Kartal
NFR-09	Reliability	Result finalization (locking scores and awarding points) is atomic: either all points are written or none; finalized results cannot be edited.	Fikriye Zeynep Battal
NFR-10	Performance	The leaderboard and history pages load in under 1 s with 10,000+ score records, using indexes and aggregate queries.	Fikriye Zeynep Battal

6. Planned Technology Stack

Relational DBMS	PostgreSQL
Database driver (raw SQL)	psycopg2
Front-end / web UI	Flask + Jinja2 templates, HTML/CSS/JavaScript

Authentication method	Flask session-based login with /register, /login, /logout; hashed passwords; role checks on protected routes
Caching layer (optional)	Not planned initially, Redis may be added for leaderboard caching

7. Work Distribution Summary

Member	Responsibility area	Functional req. IDs	Non-functional req. IDs
Şevval Tüfekçi	Accounts, Profiles & Host Ratings	FR-01, FR-02, FR-03, FR-04	NFR-01, NFR-02
Fatuma Zehra Erdem	Game Catalog & Game Libraries	FR-05, FR-06, FR-07, FR-08	NFR-03, NFR-04
Sudenaz Hasret	Game Nights & Sessions	FR-09, FR-10, FR-11	NFR-05, NFR-06
Mahmut Kartal	Participation, Reservations & Check-in	FR-12, FR-13, FR-14, FR-15, FR-16	NFR-07, NFR-08
Fikriye Zeynep Battal	Scoring, Leaderboard & Player Reports	FR-17, FR-18, FR-19, FR-20	NFR-09, NFR-10

8. Pre-Submission Checklist

☒ The team has 5 members, and the team leader is indicated.

☒ The project topic is selected from the provided list and our team name is written in the topic–team mapping document.

☒ The project idea and scope are clearly described.

☒ The proposed feature set is complex enough to require a database with at least 10 related tables.

☒ Functional and non-functional requirements are listed, and each one has exactly one responsible member.

☒ Responsibilities are balanced across members (4 functional and 2 non-functional requirements each).

☒ The primary database is a relational DBMS other than SQLite.

☒ No ORM or auto code generator will be used; all queries will be written in raw SQL.

☒ Authentication (login/register and protected routes) and user roles are planned.

☒ All gray guidance text and example rows have been removed.

For Instructor Use Only

Decision	☐Approved ☐Approved with minor changes ☐Revision required
Comments	
Reviewed by / date	